

GeoEnergy Intelligence AI

Deep Analyst Playbook & Engine Training Guide

A complete reference for building, expanding, and training the GeoEnergy Intelligence AI analysis engine — covering analytical frameworks, sector coverage, geopolitical risk methodology, AI/LLM upgrade paths, RAG architecture, and production-ready CLI prompt templates.

Project	GeoEnergy Intelligence AI
Author	Chigurupati Prabhath
Stack	React + Vite · Python FastAPI · Live API Integrations · AI-Ready Analysis Layer
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Status	Active Development — Deterministic Engine → LLM Upgrade Path

1. Analyst Mindset & Core Framework

Your analysis engine is a "synthetic analyst" — it must reason like an energy market professional who has deep knowledge of commodity dynamics, geopolitical risk, and cross-asset signals. Research shows that energy market analysts need a blend of quantitative skills, market research capability, economic acumen, and risk management discipline to generate accurate, actionable insights.

1.1 The Five Analyst Principles

- **Decision-focused output:** Every analysis sentence must help a specific stakeholder decide — HEDGE, HOLD, REDUCE, WAIT, or MONITOR. Avoid purely descriptive outputs with no action implication.
- **Scenario thinking:** Always frame in Base / Bullish / Bearish — assign probabilities. Markets are probabilistic, never certain. Your engine already implements this correctly.
- **Cross-asset connectivity:** Energy prices never move in isolation. Gold, copper, freight, FX, and rates must be checked before a final thesis is formed. Safe-haven demand (gold up + crude up) is a stronger signal than crude direction alone.
- **Fact-then-interpretation discipline:** State the data point first (WTI \$78.4/bbl, +1.2%), then the cause (supply constraint, Libya output), then the implication (fuel distributors face 3-4 week cost lag).
- **Speed via regime recognition:** Experienced analysts work fast by pattern-matching current conditions to known historical regimes — supply shock, demand shock, risk-off, policy pivot, or tech-driven transition. Your analysis engine should maintain a library of these regimes.

1.2 Analytical Skills Checklist for the Engine

Skill	Description	Priority
Fundamental Analysis	Supply/demand balances, inventory levels, OPEC+ output discipline, refinery utilization	Utilisation
Geopolitical Risk Scoring	Chokepoint status, country risk events, military/political escalation, sanctions	Critical
Technical/Quantitative	Price momentum, spread analysis (Brent-WTI, crack spreads), volatility bands, historical range	Historical range
Cross-Market Correlation	Gold/silver safe-haven signals, copper industrial demand, freight as trade indicator, USD/DXY	Indicator
Scenario Modelling	Probability-weighted base/bull/bear outcomes, price range projections, regime identification	Identification
Stakeholder Impact Analysis	Who is exposed, what their cost sensitivity is, how quickly a price move reaches them	Reaches them
Consumer Pass-Through Analysis	Energy price lag, utility bill lag, grocery freight pass-through timelines	Medium
Data Quality Assessment	How many live vs mock sources, confidence calibration, caveat flagging	Medium
LLM Prompt Engineering	Structuring context injection for domain-specific LLM calls (few-shot, chain-of-thought)	Thought
RAG Retrieval Design	Building a vector knowledge base of past events to ground LLM responses	Future

2. Complete Sector & Instrument Coverage Map

Your engine currently covers core energy sectors (Crude Oil, Natural Gas, Refined Products, Power, Renewables) plus 10 cross-market signals. This section maps the full universe of sectors the engine should understand — including gaps to close — with key instruments, drivers, and stakeholder linkages for each.

2.1 Core Energy Sectors (Already in Engine)

Sector	Key Instruments	Primary Drivers	Key Stakeholders	Engine Status
Crude Oil	WTI, Brent, Dubai Brent-WTI spread	OPEC+ output Geo-risk premium Inventory levels USD strength	Energy Traders Airlines Fuel Distributors Logistics	✓ Implemented Full stakeholder & scenario logic
Natural Gas	Henry Hub EU TTF LNG Asia (JKM)	EU storage fill rate Weather / heating demand LNG export capacity Pipeline disruptions	Utilities Industrial users LNG Traders Households	✓ Implemented Storage endpoint live (EIA)
Refined Products	Diesel, Gasoline Jet Fuel, HO=F, RB=F Crack Spreads	Refinery utilisation Maintenance cycles Gulf Coast yield Crude input cost	Airlines Logistics/Trucking Fuel Retailers Agriculture	✓ Implemented Crack spread watch defined
Power & Utilities	EU spot power €/MWh US power MWh EUA carbon price	Nat. gas-for-power Nuclear availability Renewables output Demand (weather)	Utilities Manufacturing Aluminum smelting Consumers	✓ Partial Expand nuclear & EUA tracking
Renewables	Solar/wind output Innergex, ICLN ETF Grid curtailment	Policy (IRA, EU Green Deal) Solar irradiance Grid congestion Capacity pipeline	EV / Battery Chain Power Investors Grid Operators Consumers	✓ Partial Add curtailment signal

2.2 Cross-Market Signals (Already in Engine)

Signal	Instrument / Proxy	What It Tells the Engine	Correlation to Energy	Engine Status
Gold	XAU/USD, GC=F	Safe-haven demand active — geo-fear or macro shock	+0.32 to oil proxy	✓ Full logic
Silver	XAG/USD, SI=F	Hybrid safe-haven + solar industrial demand signal	+0.25 to oil proxy	✓ Full logic
Copper	HG Futures	Industrial demand barometer. Copper down = demand softening	-0.55 to EUA	✓ Demand-driven
Wheat	ZW=F, CBOT	Food-energy nexus — fertilizer (nat. gas input)	+0.42 to freight	✓ Geopolitical
Uranium	U3O8 spot, NLR	Nuclear renaissance signal — long-term structural	+0.18 to power demand	✓ Shift in LNG
Lithium	LIT ETF, carbonate spot	EV battery cost signal. Oversupply = lower EV costs	-0.65 to EU gas	✓ Mid battery deployment
Coal	Thermal/coking spot	Backup power signal. Gas spike → coal switch	+0.55 gas to EU	✓ Price driver
Carbon / ETS	EU EUA, €/tonne	Carbon price drives power mix decisions — high power cost	+0.68 power cost	✓ Gas and coal
Aluminum	LME spot, Alcoa proxy	Smelting uses 4x industry avg power — high power cost	+0.45 power cost	✓ Smelter shutdown risk. EM demand signal.
Freight / BDI	Baltic Dry Index, FFA	Bulk trade volume proxy. BDI down = demand	+0.72 to oil	✓ Contained rates (Red Sea re-routing pressure).

2.3 Sectors to Add Next (Engine Expansion Roadmap)

Based on your current codebase, the following sectors and signals are gaps that would meaningfully increase the intelligence quality of the engine and differentiate the dashboard further.

New Sector / Signal	Why Add It	Key Data Sources	Impact Level
LNG Trade Flows	LNG is the key link between US Henry Hub and EU/Asia gas prices. US LNG exports to UK, Japan, and Asian LNG demand spikes.	Argus Price, EIA US LNG Exports, Kpler Flows	High
Refinery Utilisation Rate	US refinery utilisation (EIA weekly) is the most direct driver of US gasoline supply. A drop indicates product shortage risk within 2-4 weeks.	EIA Weekly Petroleum Report, API Crude Data	High
OPEC+ Compliance Monitor	Production discipline vs target is the #1 crude oil fundamental. OPEC+ cuts are critical for oil price stability.	OPEC Monthly Reports, Reuters, Saudi Arabia, UAE, Russia, Iraq monthly	High
Currency / DXY Index	USD strength inversely correlates with commodity prices. A strong dollar suppresses oil, gold, and metals simultaneously.	Bloomberg, FX Rates, US Fed	Medium
European Carbon (EUA)	EU Emissions Trading System price drives coal-to-gas switching. A spike in carbon prices increases industrial costs.	EUA Auctions, Bloomberg, EU Emissions	Medium
Nat. Gas Storage EU Weekly	European gas storage fill % is THE key winter risk signal. Low storage in late autumn = high risk of gas shortages.	EIA, GIE (if accessible), Eurostat	High
Strategic Petroleum Reserve (US SPR)	US SPR draws / refills directly affect WTI supply balance. A draw of 1 million barrels can suppress crude prices by 10-15%.	EIA Weekly SPR Data (2022), US Energy	High
Asian Energy Demand (China PMI)	China's 14% of global oil demand. Manufacturing PMI is a leading indicator for oil demand. A rise in China's PMI signals increased oil demand.	NBS China PMI, Reuters, China's PMIs	High
Climate / Weather Signals	Temperature anomalies drive gas-for-heating demand. A cold snap in Europe or a drought in the US can impact energy markets.	NASA, NOAA, Weather.com, AP Live	Medium
Insurance / War Risk Premiums	Red Sea / Gulf shipping insurance costs are a direct marker of geopolitical risk. A sharp increase in rates up = shipping stress.	Marsh, Lloyd's, War Risk Insurance	Medium

3. Geopolitical Risk Analysis Methodology

Geopolitical risk (GPR) has a statistically significant, positive influence on energy market vulnerability — and the effects are more pronounced during conflict situations and at high-frequency intervals. The engine must score, classify, and propagate geo-risk signals systematically. In 2023, the world's five most critical maritime chokepoints carried 71.3 million bpd of oil and about 26 Bcfd of LNG — any disruption at these points triggers acute price volatility.

3.1 The Five Critical Chokepoints

Chokepoint	Volume at Risk	Disruption Impact	Risk Premium on Brent	Watch Signal
Strait of Hormuz	~20% global crude (~17M bpd)	\$15–25/bbl spike potential on closure. Affects Middle East exports directly.	+\$4–7/bbl geopolitical base premium active	Tanker incident reports, IRGC activity, Iran sanctions news
Red Sea / Bab el-Mandeb	Asia-Europe container routes, 10–12% LNG	Re-routing via Cape of Good Hope +10-14 days voyage. Insurance costs up \$500–1,000/TEU.	+\$2–4/bbl crude freight premium when disrupted	Houthi attack reports, maritime security advisories, BDI divergence
Strait of Malacca	~25% global oil transit including Asia LNG flows	China, Japan, Korea LNG supply security risk. Extended closure would be critical.	+\$5–10/bbl regional crisis premium (rare but extreme)	ASEAN maritime security, piracy reports, LNG spot JKM spike
Suez Canal	Oil + refined product trade, ~9% global cargo (by value)	Closure = Cape re-routing adds 15-20 days to Europe-Asia trips.	+\$3–6/bbl depending on closure duration and season	Egyptian political stability, drought (canal depth), tanker grounding
Bosporus / Turkish Straits	Russian + Caspian crude (~3M bpd) to Mediterranean	Caps on Russian exports. Black Sea cargo disruption. Route for CPC crude.	+\$2–5/bbl regional impact on Urals/Mediterranean discount widening	Turkish foreign policy, Russia sanctions status, Ukraine conflict

3.2 Geo-Risk Scoring Framework for the Engine

Your engine currently classifies geo events as Critical / High / Medium and adjusts confidence, stakeholder scores, and scenario probabilities. Below is an expanded scoring framework to make the logic more precise and transparent.

Risk Level	Score Range	Criteria / Conditions	Engine Action
Critical	8.0 – 10.0	Active military conflict, direct chokepoint closure, OPEC+ production cuts, UN Security Council sanctions	Generate immediate alerts, notify all stakeholders, initiate crisis response protocols
High	6.0 – 7.9	Elevated tension, shipping lane incidents, protest-linked disruptions, UN Security Council resolutions	Escalate to High, monitor closely, prepare contingency plans
Elevated	4.0 – 5.9	Ongoing conflict without direct supply impact, diplomatic tensions, UN Security Council discussions	Monitor closely, prepare contingency plans, update risk assessments
Monitor	1.0 – 3.9	Background risk, historical conflict without escalation, UN Security Council resolutions	Monitor closely, prepare contingency plans, update risk assessments

4. Fast Analysis Workflow — The 8-Step Engine Protocol

This is the decision tree the analysis engine should follow on every call to `generate_analysis()`. Use this as the internal logic map for the engine and as a human-readable guide when writing LLM prompts that replicate the same reasoning chain.

Step 1: Frame the focal point

- Identify the primary moving force: crude direction, geo-risk event, or cross-market signal.
- Determine the time horizon: intraday (hours), near-term (1-4 weeks), or structural (months/years).
- Note which stakeholders are most directly exposed given the focal point.
- Your engine starts here: `crude_dir`, `gas_dir`, `critical_geo` → these are the frame.

Step 2: Scan core energy prices

- Check WTI, Brent, NatGas, Diesel, and Power prices and their directions.
- Calculate or estimate the Brent-WTI spread (normally \$2–4/bbl; widening = quality or logistics signal).
- Note crack spreads: a diesel crack > \$35/bbl = refinery margin stress and logistics cost pressure.
- Flag regime: is WTI above or below \$85/bbl (bullish/bearish line for airline and logistics costs)?

Step 3: Cross-check all signals

- Gold up + crude up = geo-premium confirmed (not just supply data).
- Copper down = industrial demand softness → demand-side bear case for crude is valid.
- BDI down = bulk trade slowing → secondary demand softness signal.
- Freight up + BDI down = divergence (container re-routing vs bulk demand weakness).
- USD strong = headwind for all commodity prices — note but do not overweight short-term.

Step 4: Geo-risk classification

- Classify each event: Critical / High / Elevated / Monitor (use Section 3 framework).
- Map events to chokepoints: is Hormuz, Red Sea, Bosporus, or Malacca directly affected?
- Estimate disruption probability and duration: random incident vs sustained military activity.
- Note which OPEC+ members are affected — Saudi, UAE, Iraq, Russia have the most supply weight.

Step 5: Build the three scenarios

- Base case: current trends continue; OPEC+ holds output; no new geo escalation.
- Bullish case: supply shock scenario — Hormuz incident, Libya/Nigeria shutdown, surprise cut.
- Bearish case: demand shock — China miss, US recession signals, OPEC+ surge.
- Assign probabilities: Base 50-60%, Bull 20-30%, Bear 15-25%. Adjust for geo-risk count.
- Name benefits and suffers clearly — your `SCENARIO_DEF` already does this correctly.

Step 6: Stakeholder scoring

- Start from base scores (`STAKEHOLDER_DEF`) and adjust for live direction signals.
- Key adjustments: crude up → traders +5, airlines +5, logistics +5; gas down → utilities -4.
- Apply geo-risk multiplier: each critical event = +3 for all exposed stakeholders.
- Assign posture: HEDGE (score ≥ 80), HOLD (≥ 65), MONITOR (≥ 45), WAIT/REDUCE (< 45).
- Write a one-line driver note with a specific number: "Brent \$82.1 — each \$10/bbl = ~1-2% fuel cost change."

Step 7: Synthesize the thesis

- Lead with crude direction and price level — most users care about this first.
- Add the geo-risk sentence if critical events are active: name the region and supply implication.
- Add safe-haven confirmation if gold is elevated — it validates the risk premium.

- Close with stakeholder implication: "fuel-intensive businesses face 3–4 week cost lag."
- Keep the thesis to 2–4 sentences maximum — long thesis = lower readability and clarity.
- Your engine builds this dynamically via `t_parts` — the pattern is already good.

Step 8: Data quality and confidence

- Report how many sources are live vs mock: `data_sources / 3`.
- State which sources are offline and what impact this has on confidence.
- Use your current confidence formula: $78 + (\text{live_sources} * 4) + (\text{critical_geo} * 2) + (\text{gold_up} * 3)$.
- Add a one-sentence explainability note: "This analysis is rule-based signal fusion, not generative AI."
- Flag if key signals are missing: e.g., "EU storage API offline — nat gas analysis uses cached data."

5. AI Upgrade Path: Prompting, RAG & Fine-Tuning

Your engine is deterministic today. The upgrade path to a fully AI-powered briefing system has three stages. Research shows that domain-specific prompt engineering significantly enhances LLM performance for financial tasks — and RAG pipelines combining retrieval with generation are the most practical approach for real-time energy market analysis.

5.1 Stage 1 — Prompt Engineering (Immediate, No Training Needed)

Your FastAPI already has a `/api/ai/analyze` endpoint (`ai_adapter.analyze`). The fastest upgrade is to add a structured prompt layer that passes your engine's JSON output to an LLM and asks it to rewrite or enrich the thesis, environment, and stakeholder notes.

System Prompt Template (for thesis generation):

```
You are GEI-Analyst, an energy market intelligence AI. You synthesize live price data, geopolitical risk events, and cross-market signals into concise, decision-focused briefings for traders, businesses, and consumers. You always: 1. Lead with crude oil direction and price level. 2. Reference at least one active geo-risk event if present. 3. Include a safe-haven confirmation or denial (gold direction). 4. End with a stakeholder cost implication and time lag. 5. Keep the thesis to 2-4 sentences. Return a JSON object only.
```

Context Injection Template (user message):

```
LIVE DATA SNAPSHOT: - WTI Crude: ${{wti_price}}/bbl ({{crude_dir}}), Brent: ${{brent_price}}/bbl - Natural Gas (Henry Hub): ${{gas_price}}/MMBtu ({{gas_dir}}) - Gold: {{gold_status}} | Copper: {{copper_dir}} | BDI (Freight): {{freight_dir}} - Critical Geo Events: {{critical_geo_count}} ({{critical_geo_names}}) - High-Impact News: {{high_news_count}} items - Data Sources Live: {{data_sources}}/3 CURRENT ENGINE OUTPUT (use as baseline, improve the thesis and stakeholder notes): {{engine_json}} OUTPUT: Return JSON with keys: thesis, environment, stakeholders (array), themeChips (array), confidence (int). Do not change numeric scores. Only improve text fields.
```

5.2 Stage 2 — RAG Pipeline (Medium Term)

RAG (Retrieval-Augmented Generation) combines a vector knowledge base with LLM generation. For your engine, this means storing past energy market events (with structured tags) in a vector DB, retrieving the most similar historical cases at runtime, and passing them as context to the LLM so it can reason with historical precedent rather than relying on general training data alone.

- Step 1 — Build a knowledge base:** For each major historical event (2022 Russia gas shock, 2019 Abqaiq attack, 2020 COVID demand crash, 2023 Red Sea disruption), store a structured record: `{date, event_type, prices_at_time, geo_severity, sectors_impacted, thesis_outcome, price_move_after}`.
- Step 2 — Embed and index:** Use a sentence-transformer model (e.g., `all-MiniLM-L6-v2`, free) to embed each event record into a vector. Store in ChromaDB or PostgreSQL+pgvector (both free, open-source). Your FastAPI can host the retrieval endpoint.
- Step 3 — Retrieve at runtime:** When `generate_analysis()` runs, also run a similarity search: "find 3 historical events most similar to current geo-risk + price regime." Pass retrieved events as few-shot examples in the LLM prompt.
- Step 4 — Ground the generation:** LLM uses retrieved events as analogies: "In 2019 (Abqaiq), a similar Critical geo event drove Brent +10% within 72h. Current conditions suggest a comparable +7-12% scenario in the bullish case."
- Stack recommendation for your FastAPI:** LangChain + ChromaDB (pip install `langchain chromadb sentence-transformers`). Expose as a new POST `/api/ai/rag-briefing` endpoint alongside your existing deterministic engine.

5.3 Stage 3 — Fine-Tuning (Future, When Data Exists)

- Fine-tuning requires ~50–200 curated input/output pairs: `{data snapshot → ideal briefing text}`.
- Build these pairs by running your engine on historical data and manually improving the thesis and stakeholder notes for 50+ events.

- Fine-tune a small open model (Mistral 7B, LLaMA 3.1 8B) on your curated pairs — it learns your tone, structure, and energy-specific terminology.
- Domain-specific pre-training (like FinBERT for finance) delivers the best precision for financial sentiment tasks — your energy equivalent would be "GEI-Analyst-7B".
- This is a long-term goal. Focus on prompt engineering and RAG first — they deliver 80% of the quality improvement at 10% of the cost.

6. CLI Prompt Templates for the Analysis Engine

These are ready-to-use prompts you can feed to a CLI (e.g., curl, HTTPie, custom Python CLI) or integrate into your `ai_adapter.py` to call Gemini, Claude, or OpenAI. Each prompt is designed to produce output that is compatible with your existing JSON schema from `analysis_engine.py`.

Prompt 1 — Full Executive Briefing Generation

ROLE: You are GEI-Analyst, an energy market intelligence engine. TASK: Generate a structured AI intelligence briefing for a live energy market snapshot. SNAPSHOT: {snapshot_json} RULES: 1. thesis: 2-4 sentences. Lead with crude price/direction. Include geo-risk if critical. End with stakeholder cost implication and time lag (weeks). 2. themeChips: 4-8 short tags (e.g., "Crude Bullish", "Geo-Risk Elevated", "Safe-Haven Active"). 3. environmentExplanation: 1-2 sentences on macro backdrop and data quality. 4. stakeholderImpacts: For each stakeholder, provide score (0-100 int), severity (Critical/High/Moderate/Low), posture (HEDGE/HOLD/MONITOR/WAIT/REDUCE), and a one-line note with a specific number. 5. confidence: integer 0-96. Deduct for missing data sources. OUTPUT: Return valid JSON only. No prose outside the JSON object.

Prompt 2 — Single Commodity Deep-Dive

ROLE: Energy commodity analyst with 15 years specialising in {commodity_name} markets. TASK: Analyse the current state of {commodity_name} based on the data below. DATA: - Current price: {price} {unit} - Direction vs previous session: {direction} ({change_pct}%) - Related geo-risk events: {geo_events} - Related headlines: {headlines} - Cross-market signals: {cross_signals} OUTPUT: JSON with keys: stance: one short phrase (e.g., "Risk Elevated – Geo Premium Active") thesis: 2-3 sentences drivers: array of {name, label, description} (3-5 items) stakeholders: array of {name, score, posture, note} (4-6 items) watchlist: array of {trigger, why, who} (3-5 items) consumerImpact: one paragraph Return JSON only.

Prompt 3 — Geo-Risk Event Impact Assessment

ROLE: Geopolitical risk analyst specialising in energy market impact assessment. TASK: Assess the market impact of the following geo-risk event. EVENT: - Location: {country_or_area} - Event type: {event_type} - Risk level: {risk_level} ({risk_score}/10) - Affected sectors: {affected_sectors} - Chokepoint relevance: {chokepoint} CURRENT PRICES: {price_snapshot} OUTPUT: JSON with keys: immediateImpact: one sentence (what happens to prices in 24-72h if confirmed) sustainedImpact: one sentence (what happens if event persists >2 weeks) priceRangeShift: {commodity, baseline, upside, downside} mostExposed: array of stakeholder names (top 3) invalidationCondition: one sentence (what would make this event not matter) watchTrigger: what to monitor for escalation or de-escalation Return JSON only.

Prompt 4 — Headline Classification & Why-It-Matters

ROLE: Energy intelligence analyst. Classify and contextualise the following headline. HEADLINE: "{headline_text}" CLASSIFY AND EXPLAIN: 1. sector: which primary energy sector is affected (Crude Oil / Natural Gas / Refined Products / Power / Renewables / Cross-Market) 2. impact: High Impact / Medium Impact / Low Impact 3. sentimentEffect: Bullish / Bearish / Volatile / Neutral 4. relatedRegions: array of affected geographies 5. whyItMatters: 1-2 sentences explaining the market implication 6. marketReadThrough: 1 sentence on what price or spread to watch as confirmation 7. affectedStakeholders: array of stakeholder types most exposed Return JSON only. Be concise and specific.

Prompt 5 — Rapid Scenario Refresh

ROLE: Energy market risk strategist. TASK: Update scenario probabilities and price ranges based on new data. CURRENT BASE SCENARIOS: {current_scenarios_json} NEW INFORMATION: - Critical geo events active: {critical_count} - Crude direction: {crude_dir} ({wti_price}/bbl) - Gold safe-haven: {gold_status} - Copper: {copper_dir} RULES: - Base case probability range: 40-65% - Bull case: max 45%, increase by 5% per critical geo event - Bear case: max 35%, increase if copper is down or gold

is neutral - Probabilities must sum to 100 OUTPUT: Return JSON array of 3 scenarios with updated prob, conditions, price_range, benefits, suffers, note. No other text.

7. Complete AI Intelligence Briefing — JSON Schema

This schema maps exactly to what your `analysis_engine.py` already returns and defines the contract between the deterministic engine and the future LLM layer. Any LLM integration must produce output conforming to this schema so it is a drop-in replacement for the current engine output.

```
{ "thesis": "string - 2-4 sentence market narrative", "confidence": "integer 0-96", "themeChips":
[ "string - 4-8 short regime tags", "environmentExplanation": "string - 1-2 sentence macro backdrop",
"explainabilityNote": "string - how analysis was derived, data sources used", "generatedAt": "ISO
8601 UTC timestamp", "stakeholderImpacts": [{ "id": "string
(traders|airlines|logistics|fuel-dist|utilities|investors|mfg|agri|consumers|ev)", "name": "string",
"score": "integer 0-97", "severity": "Critical|High|Moderate|Low", "posture":
"HEDGE|HOLD|MONITOR|WAIT|REDUCE", "note": "string - one line with a specific number" }],
"holderGuidance": [{ "id": "string
(crude|gas|refined|power|renewables|uranium|gold|silver|copper|freight)", "name": "string",
"commodity": "string", "posture": "BUY|HOLD|REDUCE|HEDGE|WAIT|MONITOR", "conviction":
"High|Moderate|Low", "why": "string", "invalidator": "string - what would reverse this posture",
"watch": "string - specific indicator or trigger" }], "sectorMatrix": [{ "sector": "string",
"commodities": "string (comma separated)", "impact": "Critical|High|Moderate|Low", "costPressure":
"High|Moderate|Low", "supplyRisk": "High|Moderate|Low", "demandRisk": "High|Moderate|Low",
"posture": "HEDGE|HOLD|MONITOR|WAIT", "watch": "string" }], "consumerImpact": [{ "id": "string
(gasoline|power|grocery|travel|home)", "category": "string", "commodity": "string", "direction":
"up|down|flat", "severity": "Critical|High|Moderate|Low", "explanation": "string - specific $ or %
impact with time lag", "lag": "string (e.g., 3-4 weeks)" }], "scenarios": [{ "id":
"base|bullish|bearish", "type": "string", "prob": "integer (must sum to 100)", "color": "hex color
string", "conditions": "string - what must be true for this scenario", "price_range": "string - e.g.
Brent $78-85, Gas $2.0-2.5/MMBtu", "benefits": ["string - stakeholder/asset class that benefits"],
"suffers": ["string - stakeholder/asset class that suffers"], "note": "string - 1-2 sentence
narrative" }], "pressureMix": [{ "label": "Geo Risk|Price Action|Headlines|Cross-Market|Historical
Trend|Sector Risk", "weight": "integer (sum = 100)", "color": "hex string", "description": "string"
}], "watchlist": [{ "id": "string", "name": "string", "trigger": "string - specific level or event",
"why": "string - market implication", "who": "string - affected stakeholders" }], "dataFreshness": {
"prices": "live|mock|offline", "geoRisk": "live|mock|offline", "news": "live|mock|offline" } }
```

8. Speed Principles & Quick-Reference Decision Card

The analysis engine must deliver insights in near-real-time. These principles guide fast, high-quality analysis without sacrificing accuracy.

8.1 Speed Principles

- **Recency over completeness:** Focus on the last 24–72 hours of moves. A week-old data point does not change the thesis unless it is a storage or OPEC+ report.
- **Regime recognition shortcut:** Map current conditions to one of 5 regimes instantly — Supply Shock / Demand Shock / Risk-Off Flight / Policy Pivot / Tech Transition. Each regime has a pre-built stakeholder and scenario template.
- **Use thresholds, not trends:** Only flag a price move if it crosses a meaningful threshold (e.g., Brent > \$90, BDI < 1500, crack > \$35). Noise below thresholds = no engine response.
- **150-word thesis limit:** If the thesis is longer than 150 words, it is a report, not a briefing. Compress aggressively.
- **One number per stakeholder:** Every stakeholder note must contain one specific number — a price, a percentage, a time lag, or a dollar impact. "Brent \$82.1 — each \$10/bbl = ~\$180M annual cost for a mid-size airline."
- **Always state confidence + caveats:** Never present analysis without a confidence score and a one-sentence note about which data is live vs mock.

8.2 Regime Quick-Reference Card

Regime	Trigger Conditions	Default Thesis Direction	Key Stakeholders	Engine Posture
Supply Shock	Geo event Critical/High + crude up > 2% + gold up + freight up	Bullish crude / gas. Fuel costs escalate for businesses.	Airlines, Logistics, Fuel Dist, Traders	HEDGE traders, HEDGE airlines, Bull prob +10%
Demand Shock	Copper down + crude down + BDI down + China miss signal	Bearish crude / gas. Industrial demand weakness signal.	Crude Producers, Energy Traders, EM Exporters	REDUCE crude, REDUCE gas, Bear prob +10%
Risk-Off Flight	Gold up sharply + crude down + equities down + USD up	Mixed energy. Safe-haven active. Recession fear building.	Investors (gold), Consumers benefit from lower fuel	BUY gold, HOLD crude, Base prob unchanged
Policy Pivot	OPEC+ surprise cut or production surge announced	Direct crude move. Cut = +\$5-10/bbl. Surge = -\$5-8/bbl.	Energy Traders (all sides); Airlines & Logistics	HEDGE on cut, REDUCE on surge, Bull/Bear ±15%
Tech Transition	Lithium oversupply + renewables record + EV adoption beat	Long-term bearish for fossil fuels. Near-term neutral.	EV chain, Renewables Investors, Coal/Gas producers (negative)	WAIT lithium, HOLD renewables, Long-term REDUCE coal/gas

9. Engine Improvement Priority Roadmap

Based on a full review of analysis_engine.py and main.py, here is a prioritised list of concrete improvements — from quick fixes this week to longer-term AI layer additions.

Priority	Improvement	Why It Matters	Effort
P1 — Immediate	Add EU gas storage via GIE AGS interface. Fill rate is THE key nat gas signal. Your US EIA storage is a nice-to-have — adding EU completes the picture.	EU storage	Low (1 day)
P1 — Immediate	Add LNG export volume check (US DOE). Exports link Henry Hub to global gas prices. A disruption triggers both domestic oversupply and global shortages.	LNG exports	Low (1-2 days)
P1 — Immediate	Add Brent-WTI spread to crack log. When Brent-WTI spread widens beyond \$5, it signals logistics disruptions. Current engine uses east coast WTI.	Brent-WTI	Low (Quality issue)
P2 — Short-term	Add China Manufacturing PMI watch. China accounts for ~14% of global oil demand. PMI < 50 = demand side crude bear. Currently copper PMI only.	China PMI	Low-Medium
P2 — Short-term	Add refinery utilisation % from EIA. Crack spread analysis without utilisation context misses 50% of the picture. Below 85% = shortage risk.	Refinery util	Medium (EPA parse)
P2 — Short-term	Add USD/DXY signal to cross-market analysis. Dollar suppresses all commodity prices — it is missing from your cross-market array. Low effort.	USD/DXY	Low (1 day)
P3 — Medium-term	Add weather / temperature anomalies signal. Temperature deviations drive gas-for-heating demand. Pair with OpenWeatherMap or NOAA free.	Weather	Medium (OpenWeather)
P3 — Medium-term	Implement prompt engineering. Use Gemini/Claude/OpenAI to enrich thesis and stakeholder notes (ext. LLM). On the existing engine J.	LLM layer	Medium (ext. LLM)
P3 — Medium-term	Add OPEC+ compliance monitor. Monthly actual vs target production is the #1 crude fundamental. Build a scraper or use EIA/IEA monthly.	OPEC+ data	Medium-Build
P4 — Long-term	Build RAG knowledge base. Store 50-100 historical energy crisis events as structured data. High (new module).	Historical DB	High (new module)
P4 — Long-term	Add SPR (Strategic Petroleum Reserve) tracking. EIA SPR releases are direct price tools. Tracking levels and policy signals adds a macro policy layer.	SPR data	Medium
P4 — Long-term	Add war-risk / marine insurance premium signal. 95% of firms experienced a supply chain loss in the last year (insurance) premium spikes correlate with risk events.	War-risk	High (new data source)